

Module 3

2 Marks Questions (Study Materials):

1. Why do we need Regression?
2. State 2 advantages of regression.
3. What are dependent variables in a linear regression?
4. What are independent variables in a linear regression?
5. What is Linear Regression?
6. What is Logistic Regression?
7. Name different types of logistic regression.
8. What is indicated by m and c in equation $y=mx + c$?
9. What is regression coefficient?
10. What is multiple linear regression?
11. State two differences between linear regression and logistic regression.
12. State two differences between regression and classification.
13. State one example of application of linear regression.
14. State one example of application of logistic regression.
15. State MLE (Maximum Likelihood Estimation).
16. State the difference between likelihood and probability.
17. State one example of application of MLE (Maximum Likelihood Estimation).
18. What is MAE (Mean Absolute Error)?
19. What is outlier?
20. What is MSE (Mean Square Error)?

5 Marks Questions (Book Bank):

SHORT ANSWER-TYPE QUESTIONS (5 MARKS EACH)

1. What is a dependent variable and an independent variable in a linear equation? (5)
2. What is simple linear regression? Give one example. (5)
3. Define slope in a linear regression. (5)
4. Find the slope of the graph where the lower point on the line is represented as (-3, -2) and the higher point on the line is represented as (2, 2). (5)
5. What are the conditions of a positive slope in linear regression? (5)
6. What are the conditions of a negative slope in linear regression? (5)
7. What is multiple linear regression? (5)
8. Define sum of squares due to error in multiple linear regression. (5)
9. Define sum of squares due to regression in multiple linear regression. (5)
10. What is multicollinearity in regression equation? (5)
- ~~11. What is heteroskedasticity?~~
- ~~12. Explain ridge regression.~~
- ~~13. Explain lasso regression.~~
14. What is polynomial regression?
- ~~15. Explain basis function.~~
16. Explain logistic regression. (5)

LONG ANSWER-TYPE QUESTIONS

5 Marks Questions (Study Materials):

1. Explain the working principle of Linear Regression with example and explanation.
2. Discuss the advantages and disadvantages of Linear Regression.
3. Explain the working principle of Logistic Regression with example and explanation.
4. Discuss the advantages and disadvantages of Logistic Regression.
5. Compare the working principles of Linear regression and Random forest?
6. Compare the working principles of Linear regression and Logistic Regression.
7. What are the assumptions do a Linear Regression algorithm make about underlying dataset?
8. State 5 differences between Linear Regression and Logistic Regression with detail explanation.
9. Write a short note on MLE (Maximum Likelihood Estimation).
10. Write a short note on Regression and its Applications.

10 Marks Questions (Book Bank):

LONG ANSWER-TYPE QUESTIONS (10 MARKS EACH)

1. Define simple linear regression using a graph explaining slope and intercept. (10)
2. Explain rise, run, and slope in a graph. (10)
3. Explain slope, linear positive slope, and linear negative slope in a graph along with various conditions leading to the slope. (10)
- + 4. ~~Explain curve linear negative slope and curve linear positive slope in a graph.~~
5. Explain maximum and minimum point of curves through a graph. (10)
- + 6. ~~Explain ordinary least square with formula for a and b.~~
- + 7. ~~Explain the OLS algorithm with steps.~~
8. What is standard error of the regression? Draw a graph to represent the same. (10)
9. Explain multiple linear regression with an example. (10)
10. Explain the assumptions in regression analysis and BLUE concept. (10)
11. Explain two main problems in regression analysis. (10)
12. How to improve accuracy of the linear regression model? (10)
13. Explain polynomial regression model in detail with an example. (10)
14. Explain logistic regression in detail. (10)
15. What are the assumptions in logistic regression? (10)
16. Discuss maximum likelihood estimation in detail. (10)

10 Marks Questions (Study Materials):

1. How can linear regression be evaluated in terms of its effectiveness for predictive modelling, and what are the key factors that influence its accuracy?
2. Describe the working principles of Linear Regression and Logistic Regression with example.
3. Discuss the performance matrix and ML algorithms required for handling Imbalance data.
4. How to tackle the missing values in a dataset? What are the techniques that are required to fill the missing values?
5. Explain in detail the working methodology of MLE (Maximum Likelihood Estimation) stating the difference between likelihood and probability.